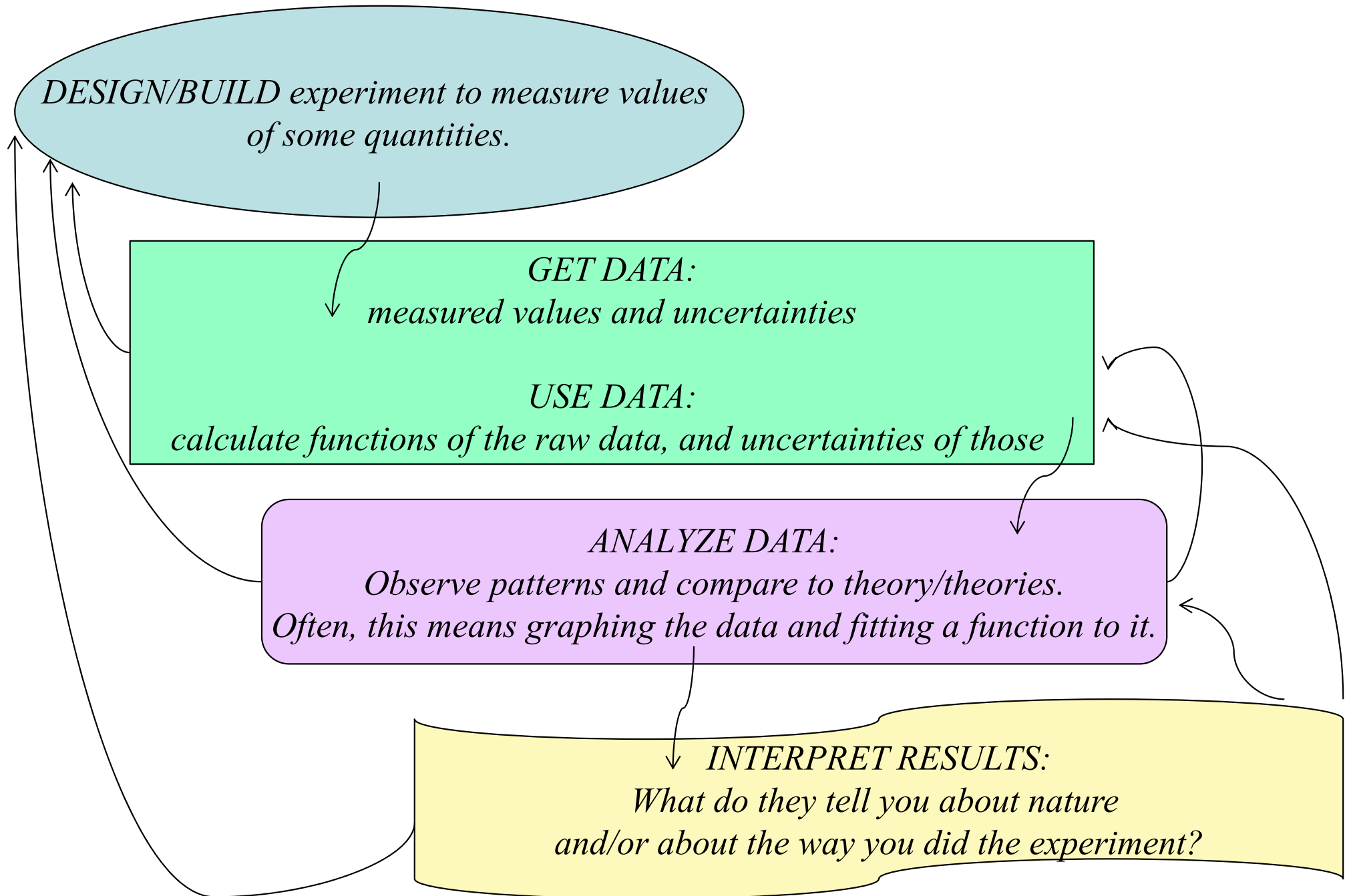


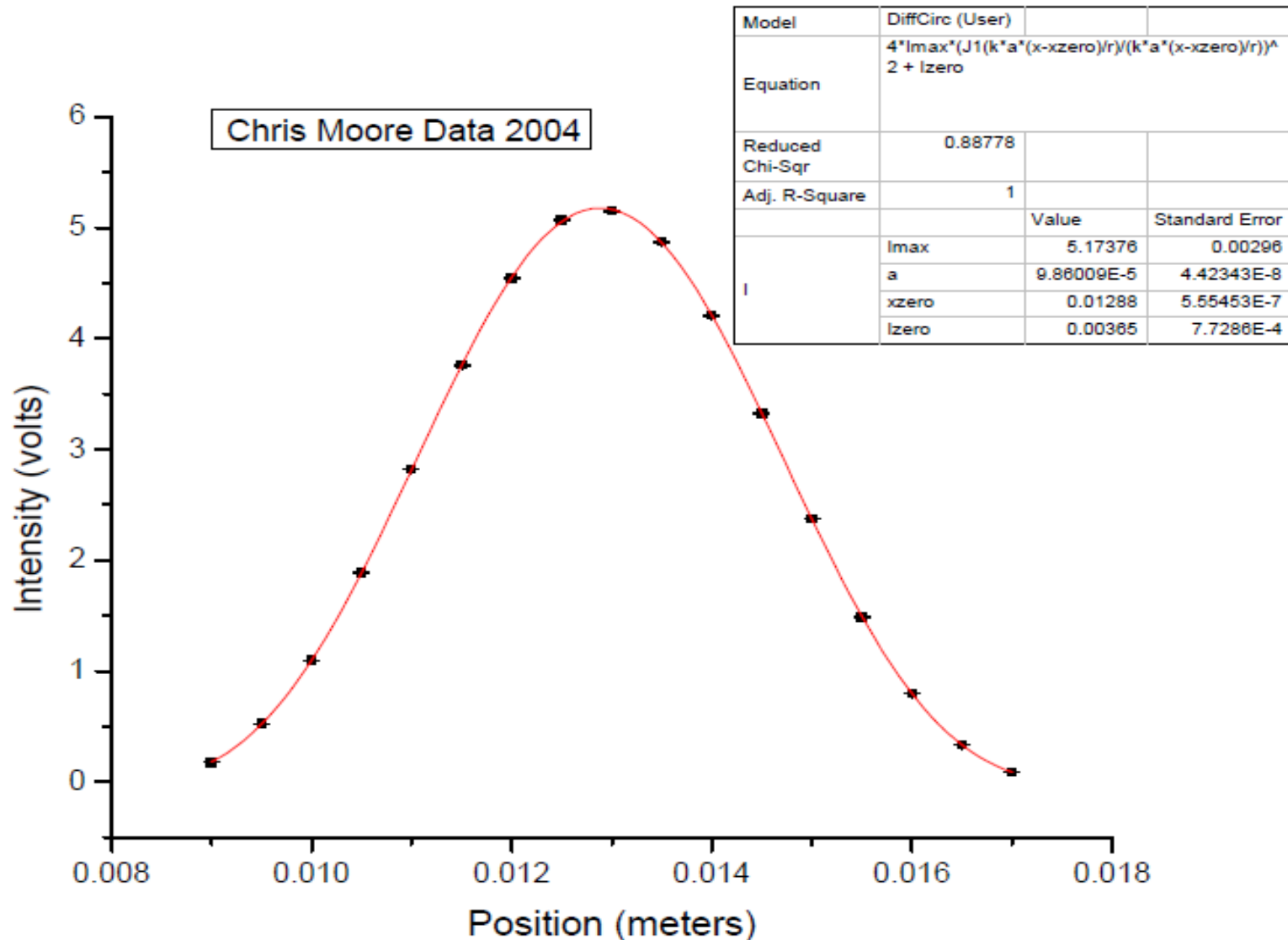
Ph134, Lecture 1

- **Why optics?**
- **Course logistics**
- **Mathematical underpinnings
of data analysis, part I**

The anatomy of an experiment

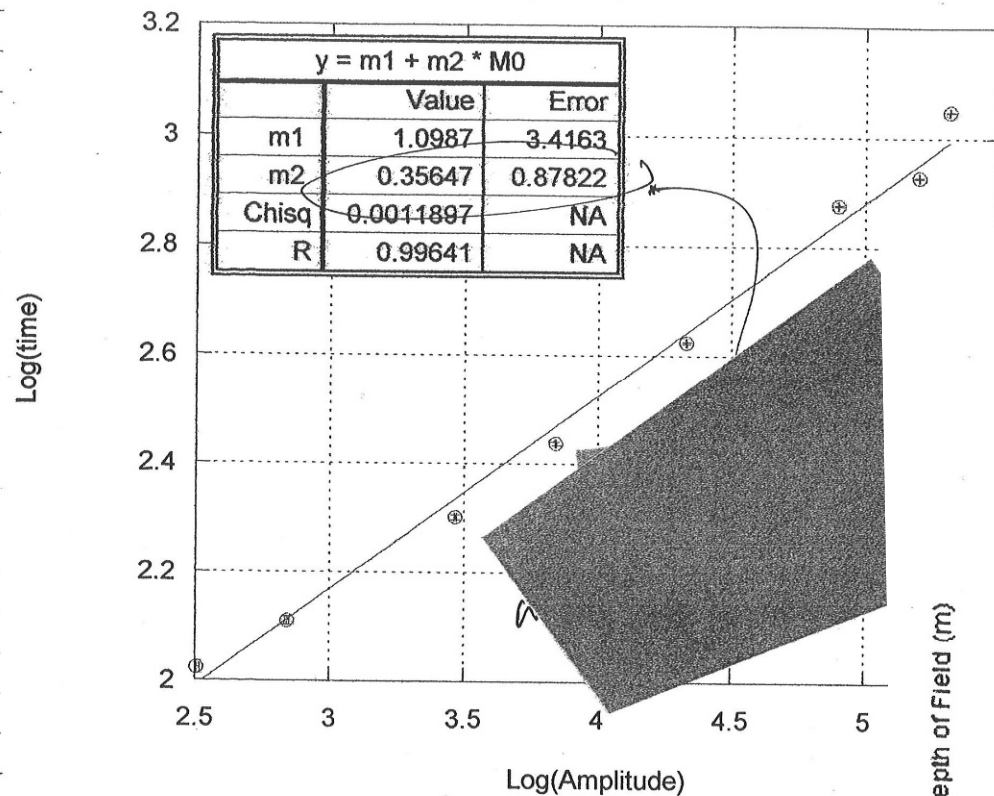


A powerful tool for analyzing data: a graph with error bars and a weighted least- squares fit to a function motivated by theory

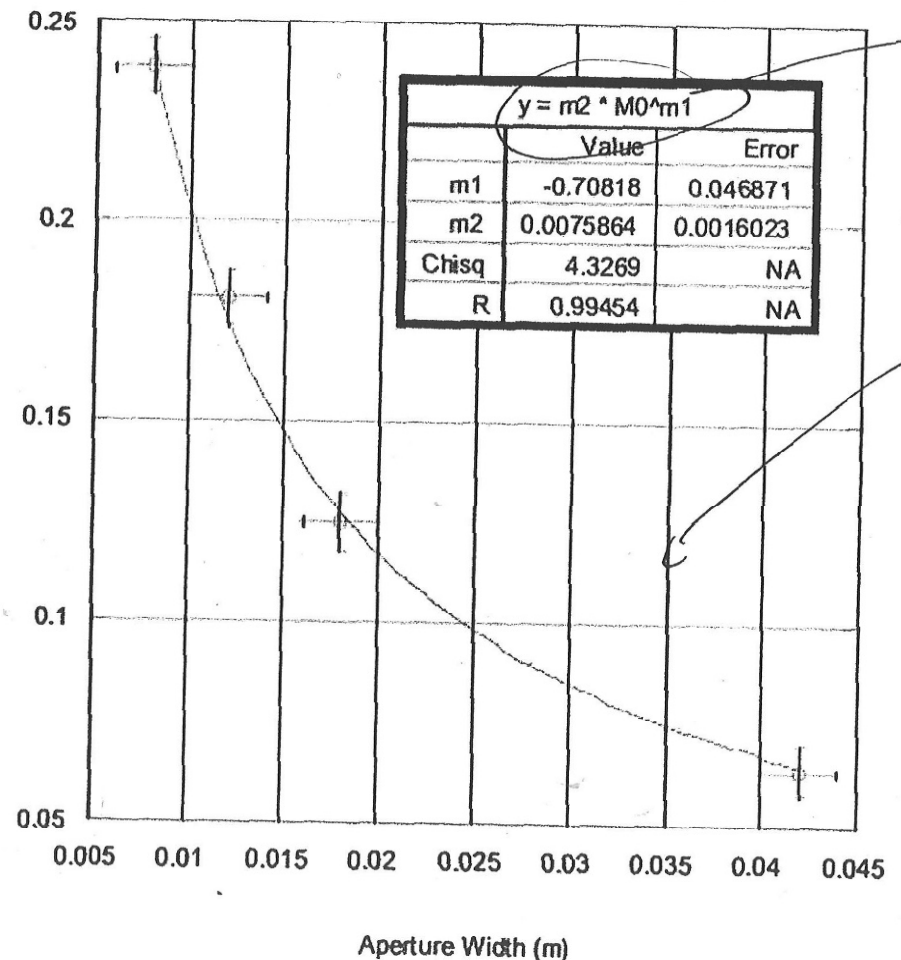


A scientist has to know how to use the tool!

Large Amplitudes



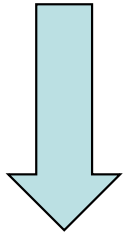
Depth of Field (m) v. Aperture Width (m)



Random Errors in the Lab

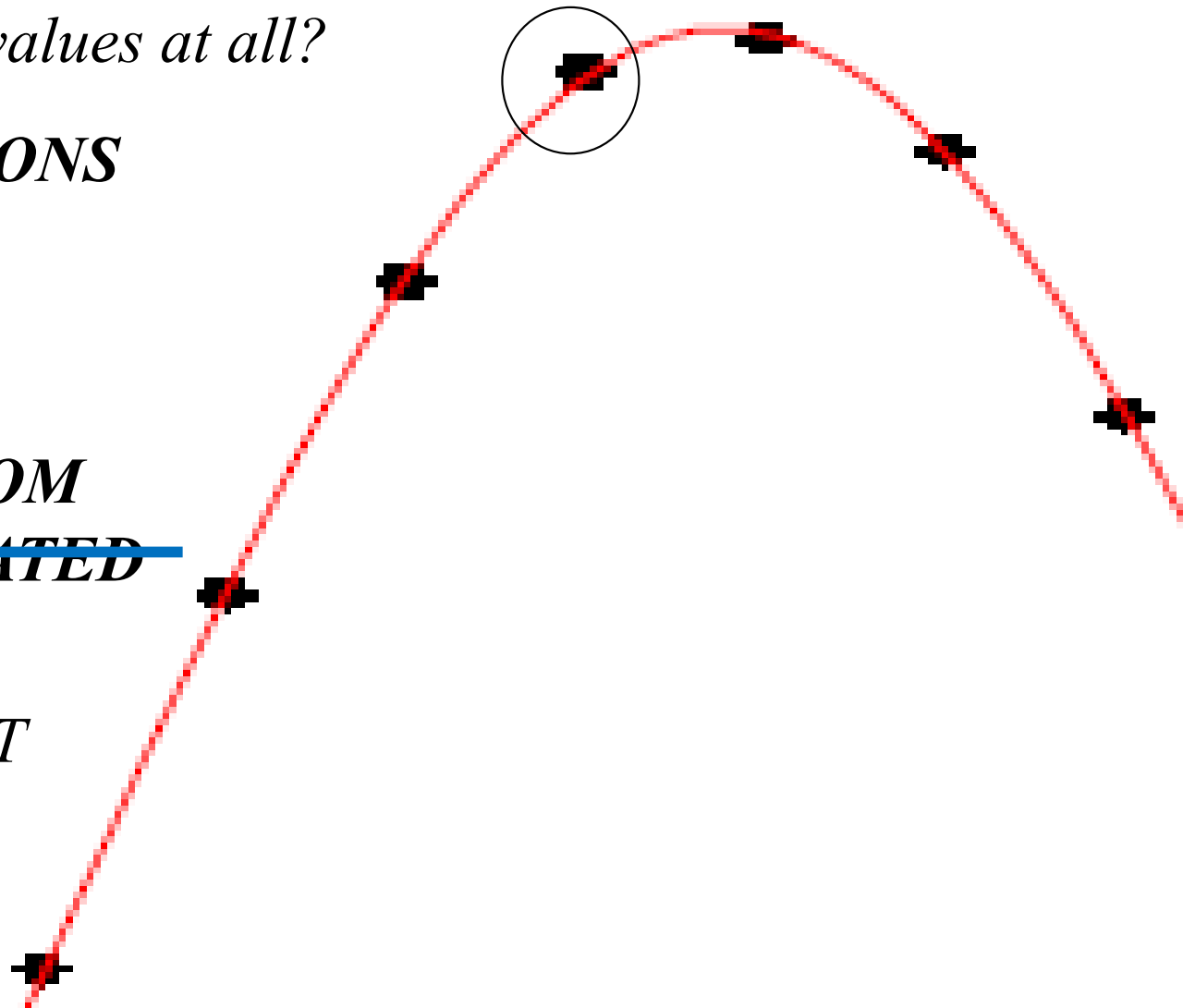
Theory says light intensity at a fixed position away from the center of an aperture should have a fixed value. Why do repeated trials give different values at all?

MANY REASONS



***MODELING RANDOM
ERROR IS ~~COMPLICATED~~
SIMPLE***

***(THE CENTRAL LIMIT
THEOREM)***



The Central Limit Theorem

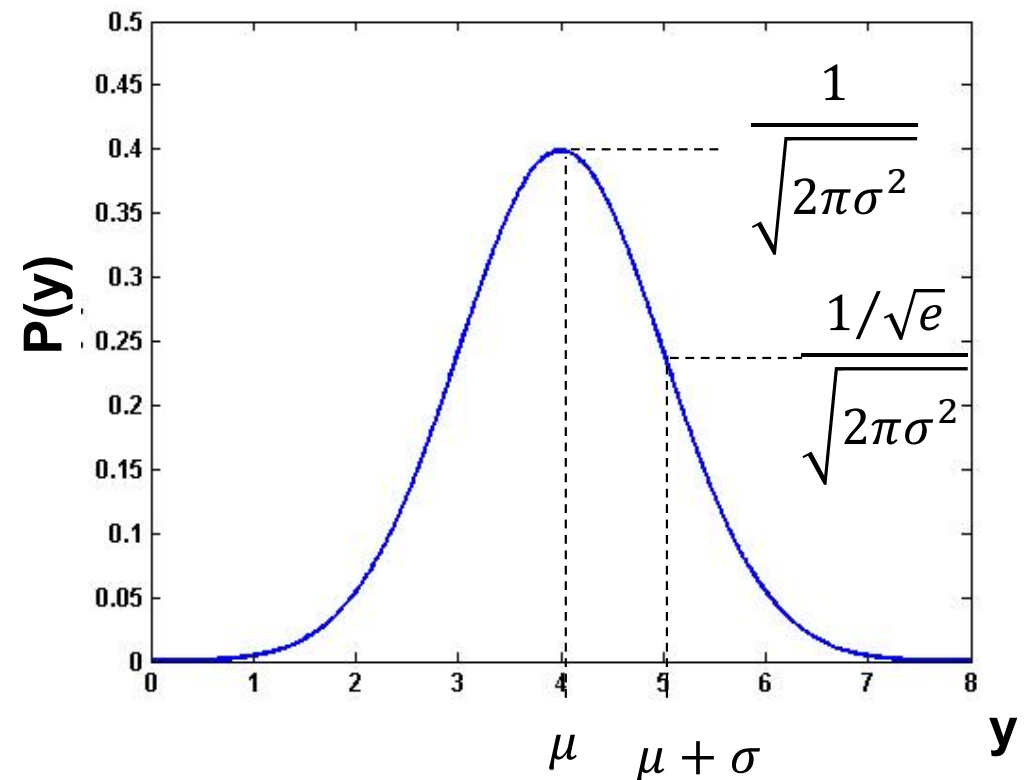
Whenever the random error in a measurement is the sum of many individual random error sources (of finite variance), the overall random error in the measurement follows a normal distribution (Gaussian):

$$P(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$$

μ = true mean of the distribution

σ = true standard deviation

σ^2 = true variance



A more specific version of the Central Limit Th'm:

Suppose $P(y)$ is a probability distribution with mean μ and finite variance σ^2 . $P(y)$ does not need to be Gaussian.

*Given N independent trials $\{y_1, y_2, \dots, y_N\}$,
i.e., N values sampled at random from the distribution $P(y)$,
let $\bar{y} = \frac{1}{N} \sum_{n=1}^N y_n$ be the mean of the N trials.*

If we perform the entire process – sampling N values and averaging them to get a $\bar{y}$ value – many times, the set of $\bar{y}$ values themselves follow some probability distribution.

As $N \rightarrow \infty$, the distribution followed by the $\bar{y}$ values approaches a Gaussian with mean μ and variance σ^2 / N .

Properties of a normal distribution (mean μ , variance σ^2):

$$\int_{-\infty}^{\infty} P(y) dy = 1$$

Note $P(y)$ is not a probability, but a probability density:

$P(y)dy$ is the probability of getting a measurement result between y and $y+dy$.

$$\langle y \rangle \equiv \int_{-\infty}^{\infty} y P(y) dy = \mu$$

$$\begin{aligned} \langle (y - \mu)^2 \rangle \\ \equiv \int_{-\infty}^{\infty} (y - \mu)^2 P(y) dy = \sigma^2 \end{aligned}$$

$$\int_{\mu-\sigma}^{\mu+\sigma} P(y) dy = 0.68$$

$$\int_{\mu-2\sigma}^{\mu+2\sigma} P(y) dy = 0.95$$

$$\int_{\mu-3\sigma}^{\mu+3\sigma} P(y) dy = 0.997$$

From N trials to a value and an uncertainty:

Task: Given N independent trials $\{y_1, y_2, \dots, y_N\}$, i.e., N values sampled at random from the distribution $P(y)$, estimate μ and attach an uncertainty to that estimate of μ .

Claimed solution (we will prove):

Sample mean

$$\bar{y} = \frac{1}{N} \sum_{n=1}^N y_n \quad \text{estimates } \mu.$$

Sample variance

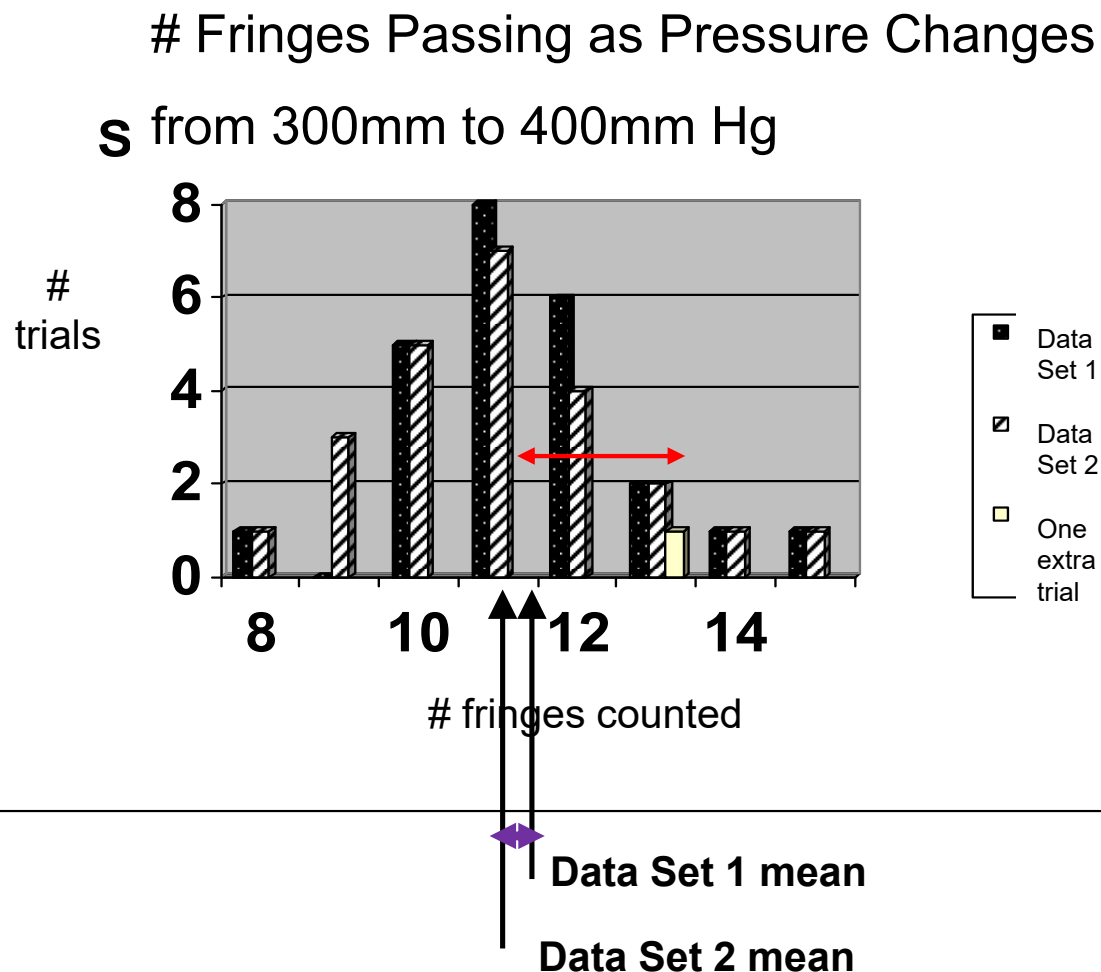
$$s^2 = \frac{1}{N-1} \sum_{n=1}^N (y_n - \bar{y})^2 \quad \text{estimates } \sigma^2.$$

Standard error of the mean

(SEM)

$$s_{\bar{y}} = \frac{s}{\sqrt{N}} \quad \begin{array}{l} \text{gives uncertainty of } \bar{y} \\ \text{as an estimate of } \mu. \end{array}$$

Disclaimer: The following data sets are completely made up to illustrate the concept of standard deviation vs. standard error. Do not expect the data of our friend I.N.Competent to resemble your own (real) data, either in spread or in mean value!



Standard deviation:

Comparable to width of either data set, or expected distance of “one extra trial” from the mean of either data set.

Standard error (of the mean):

*Comparable to deviation of either **mean** from the true mean, and thus comparable to the deviation of the Data Set 1 and Data Set 2 means from each other.*